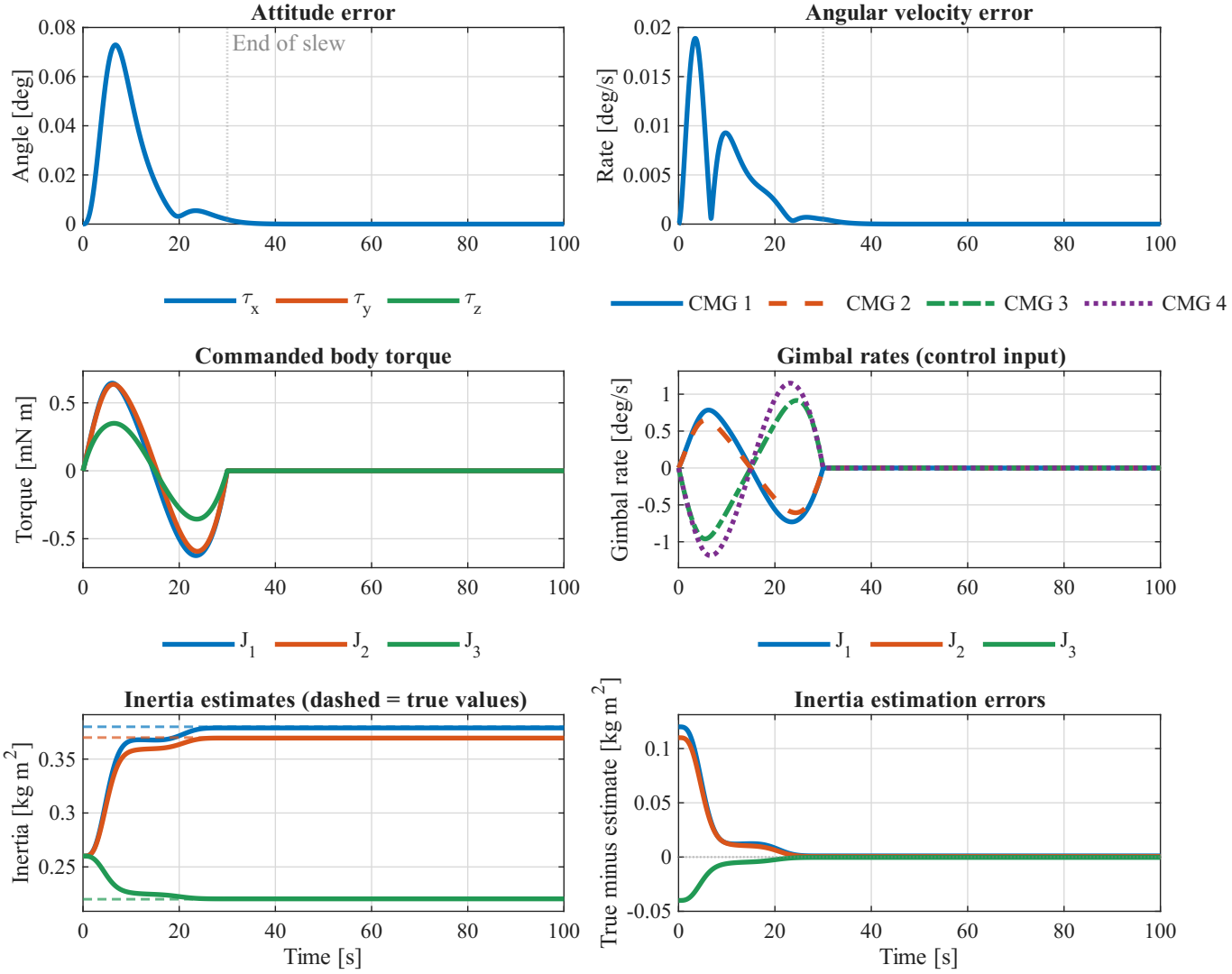

Attitude error
End of slew
Angle [deg]
0.08
0.06
0.04
0.02
0
0
20
40
60
80
100
Angular velocity error
Rate [deg/s]
0.02
0.015
0.01
0.005
0
τx
τy
τz
CMG 1
CMG 2
CMG 3
CMG 4
Commanded body torque
Torque [mN m]
0.5
0
-0.5
Gimbal rates (control input)
Gimbal rate [deg/s]
1
0.5
0
-0.5
-1
J1
J2
J3
Inertia estimates (dashed = true values)
Inertia [kg m2]
0.35
0.3
0.25
Time [s]
Inertia estimation errors
True minus estimate [kg m2]
0.1
0.05
0
-0.05
Time [s]